

- RAG Learning

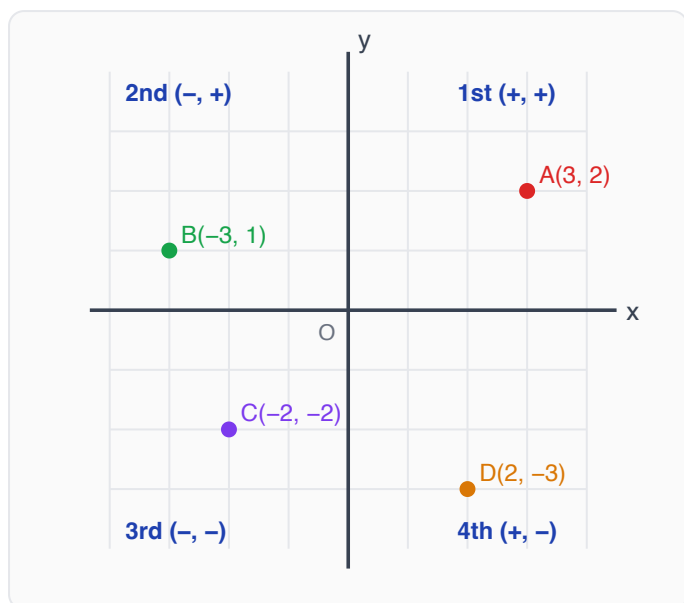
2.3 Linear Graphs & Coordinates

Pearson Edexcel GCSE Mathematics 1MA1 — Algebra — spec refs A8–A10, G11 — Papers 1 (non-calculator), 2 and 3 (calculator)

This topic is about locating points anywhere on a coordinate grid (A8), drawing straight-line graphs and finding their equations, including lines that are parallel (A9), reading and interpreting gradients and intercepts both from graphs and from equations (A10), and solving geometry problems set on coordinate axes (G11). For 1MA1 the only Higher-tier extra is using $y = mx + c$ to recognise and construct **perpendicular** lines (the Higher wording of A9).

2.3.1 Coordinates in All Four Quadrants (A8)

Every point on a grid is described by an ordered pair (x, y) . The first number is the **horizontal** position (positive to the right of the origin, negative to the left); the second is the **vertical** position (positive above, negative below). The two axes meet at the **origin** $O(0, 0)$ and split the grid into four **quadrants**, numbered anticlockwise from the top right.



Exam Tip: Read across first, then up or down. Edexcel often puts the axes in the middle of the grid, so check the sign of each coordinate: a point in the 2nd quadrant has a negative x and a positive y .

Midpoint of a Line Segment

For the points (x_1, y_1) and (x_2, y_2) ,

$$\text{midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Each coordinate of the midpoint is the **average** of the two matching coordinates.

Worked Example: Work out the midpoint of $P(-6, 5)$ and $Q(2, -3)$.

$$x = \frac{-6 + 2}{2} = -2, \quad y = \frac{5 + (-3)}{2} = 1$$

$$\text{Midpoint} = (-2, 1)$$

Worked Example: A is $(3, -2)$ and the midpoint of AB is $M(-1, 2)$. Work out the coordinates of B .

From A to M is -4 across and $+4$ up. Repeat the same step from M :

$$B = (-1 - 4, 2 + 4) = (-5, 6) \quad (\text{check: } 2 \times (-1) - 3 = -5 \text{ and } 2 \times 2 - (-2) = 6)$$

2.3.2 Geometrical Problems on Coordinate Axes (G11)

G11 questions put a shape on a grid and ask you to find a missing vertex, a length, an area or a property. Use what you know about the shape. In a parallelogram $ABCD$, the movement from A to B is identical to the movement from D to C ; in a rectangle or square, horizontal and vertical sides can be read straight off the grid.

Worked Example: $A(-2, 1)$, $B(3, 2)$ and $C(5, 6)$ are three vertices of parallelogram $ABCD$. Find D .

The step from B to C is $(+2, +4)$, so the step from A to D is the same:

$$D = (-2 + 2, 1 + 4) = (0, 5)$$

Worked Example: The lines $y = 1$, $x = 7$ and $y = x + 1$ enclose a triangle. Work out its area.

$y = x + 1$ meets $y = 1$ at $(0, 1)$ and meets $x = 7$ at $(7, 8)$. The lines $x = 7$ and $y = 1$ meet at $(7, 1)$.

The triangle has a right angle at $(7, 1)$ with base 7 and height 7: area = $\frac{1}{2} \times 7 \times 7 = 24.5$ square units.

Exam Tip: Sketch the points on a quick grid even when no diagram is given. A sketch shows you which sides are horizontal or vertical and catches answers that are obviously in the wrong place.

2.3.3 Plotting Straight-Line Graphs (A9)

An equation in which x and y appear only to the power 1 (no x^2 , no $\frac{1}{x}$) gives a **straight line**. To draw it, complete a **table of values**, plot the points accurately and join them with a single ruled line.

x	-1	0	1	2	3
$y = 4 - 3x$	7	4	1	-2	-5

- The y -values change by the same amount each time (-3 here). A value that breaks the pattern is an arithmetic slip.
- Three points are enough; any more act as checks.
- Extend the line across the full range of x -values given in the question.

Lines You Should Recognise

Equation	What it looks like
$y = k$ (e.g. $y = -2$)	Horizontal line through $(0, k)$; gradient 0
$x = k$ (e.g. $x = 5$)	Vertical line through $(k, 0)$
$y = 0$ and $x = 0$	The x -axis and the y -axis
$y = x$	Through the origin at 45° , sloping up
$y = -x$	Through the origin at 45° , sloping down

Equations in the Form $ax + by = c$

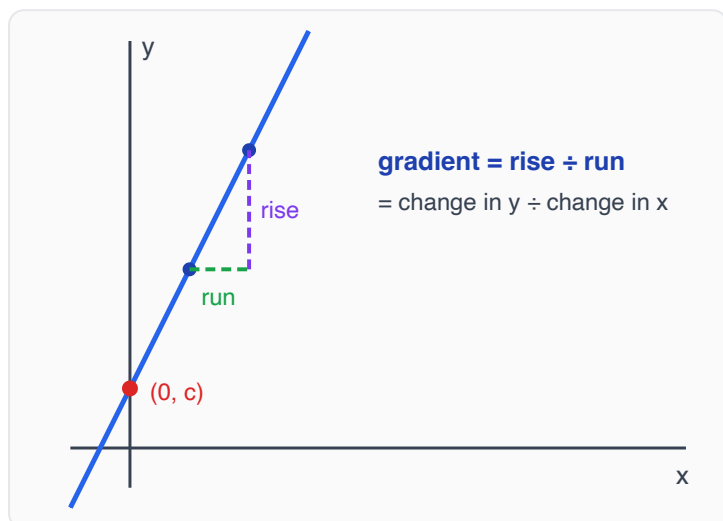
Find where the line meets each axis. For $3x + 4y = 24$: setting $x = 0$ gives $4y = 24$, so $y = 6$; setting $y = 0$ gives $3x = 24$, so $x = 8$. Plot $(0, 6)$ and $(8, 0)$ and join them.

2.3.4 Gradients and Intercepts: $y = mx + c$ (A9, A10)

Any straight line that is not vertical has an equation of the form

$$y = mx + c$$

The number m is the **gradient**: how much y increases when x increases by 1. The number c is the **y -intercept**: the line crosses the y -axis at $(0, c)$.



Gradient from two points (x_1, y_1) and (x_2, y_2) :

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{change in } y}{\text{change in } x}$$

- Uphill from left to right: positive gradient. Downhill: negative gradient.
- A horizontal line has gradient 0.
- A larger value of $|m|$ means a steeper line.

Worked Example: Work out the gradient of the line through $(-3, 2)$ and $(5, -4)$.

$$m = \frac{-4 - 2}{5 - (-3)} = \frac{-6}{8} = -\frac{3}{4}$$

Reading m and c from an Equation

Make y the subject first; then the gradient is the number multiplying x and the intercept is the constant term.

Equation	As $y = mx + c$	Gradient	y -intercept
$y = 9 - 4x$	$y = -4x + 9$	-4	$(0, 9)$
$3y = 12x - 6$	$y = 4x - 2$	4	$(0, -2)$
$5x + y = 3$	$y = -5x + 3$	-5	$(0, 3)$
$3x + 6y = 12$	$y = -\frac{1}{2}x + 2$	$-\frac{1}{2}$	$(0, 2)$

Exam Tip: "Write down the gradient of $y = 9 - 4x$ " is a one-mark question that many students drop by answering 9. The gradient is the coefficient of x , whichever order the terms are in.

Where a Line Crosses the x -axis

On the x -axis, $y = 0$. For $y = 3x - 12$: $0 = 3x - 12$ gives $x = 4$, so the line crosses at $(4, 0)$.

2.3.5 Finding the Equation of a Line (A9)

One Point and the Gradient

1. Put the gradient into $y = mx + c$.
2. Substitute the x and y of the point.
3. Solve for c and write out the full equation.

Worked Example: Find an equation of the line with gradient -2 that passes through $(3, 1)$.

$$1 = -2 \times 3 + c \Rightarrow 1 = -6 + c \Rightarrow c = 7$$

$$y = -2x + 7$$

Two Points

1. Work out the gradient from the two points.
2. Substitute either point to find c .
3. Check that the other point fits.

Worked Example: Find an equation of the line through $(-2, -7)$ and $(3, 8)$.

$$m = \frac{8 - (-7)}{3 - (-2)} = \frac{15}{5} = 3$$

$$-7 = 3 \times (-2) + c \Rightarrow c = -1, \text{ so } y = 3x - 1$$

$$\text{Check with } (3, 8): 3 \times 3 - 1 = 8 \checkmark$$

Worked Example: Find the line through $(-1, 4)$ and $(5, 0)$. Give your answer in the form $ax + by = c$ where a , b and c are integers.

$$m = \frac{0 - 4}{5 - (-1)} = \frac{-4}{6} = -\frac{2}{3}$$

$$4 = -\frac{2}{3}(-1) + c = \frac{2}{3} + c \Rightarrow c = \frac{10}{3}$$

$$y = -\frac{2}{3}x + \frac{10}{3} \Rightarrow 3y = -2x + 10 \Rightarrow 2x + 3y = 10$$

Parallel Lines

Parallel lines have equal gradients. For example, $y = 5x - 1$ and $2y = 10x + 7$ (which is $y = 5x + 3.5$) are parallel because both have gradient 5.

Worked Example: Find an equation of the line parallel to $y = 5x + 2$ that passes through $(-2, 3)$.

$$\text{Same gradient, so } m = 5. \quad 3 = 5 \times (-2) + c \Rightarrow c = 13$$

$$y = 5x + 13$$

Exam Tip: To decide whether lines are parallel, rearrange every equation into $y = mx + c$ before comparing. $4x + 2y = 7$ has gradient -2 , not 4 .

2.3.6 Interpreting Gradients and Intercepts in Context (A10)

When a straight-line graph models a real situation, the gradient is a **rate of change** (units of the vertical axis *per* unit of the horizontal axis) and the intercept is the value when the horizontal quantity is zero, often a fixed charge or starting amount. Reading a gradient as a rate of change also links to R14 in the Ratio strand.

Model	Gradient means	Intercept means
Electrician: $C = 30h + 55$ (£, hours)	£30 for each hour	£55 call-out charge
Draining tank: $V = 500 - 20t$ (litres, minutes)	Volume falls by 20 litres per minute	500 litres at the start
Distance–time graph	Speed	Distance from the start point at time 0
Conversion graph, pounds to dollars	Exchange rate (dollars per pound)	0: the line passes through the origin

Worked Example: An electrician charges a call-out fee plus an hourly rate. A 2-hour job costs £115 and a 6-hour job costs £235. Work out the hourly rate and the call-out fee.

$$\text{Gradient} = \frac{235 - 115}{6 - 2} = \frac{120}{4} = 30, \text{ so the hourly rate is } \mathbf{\pounds 30 \text{ per hour}}.$$

$$115 = 30 \times 2 + c \Rightarrow c = 55, \text{ so the call-out fee is } \mathbf{\pounds 55}. \text{ Model: } C = 30h + 55.$$

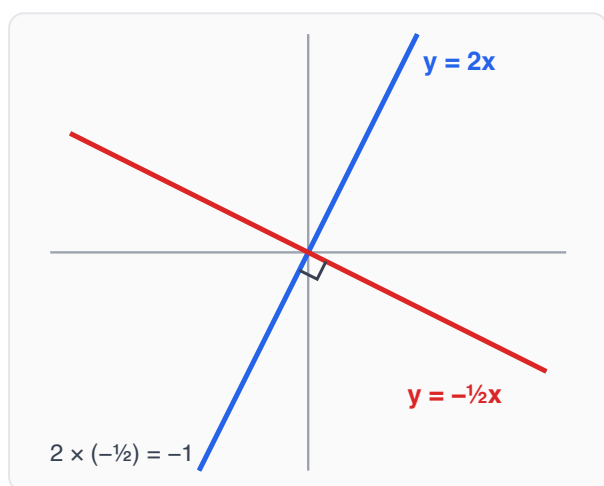
Exam Tip: “Interpret the gradient” needs a sentence with units and context, for example “the volume of water decreases by 20 litres every minute”. A bare number scores nothing.

2.3.7 Perpendicular Lines (A9) Higher

Two lines are **perpendicular** (cross at 90°) when their gradients multiply to give -1 :

$$m_1 \times m_2 = -1 \quad \text{so} \quad m_2 = -\frac{1}{m_1}$$

To get the perpendicular gradient, take the reciprocal *and* change the sign (the **negative reciprocal**).



Gradient m_1	Perpendicular gradient m_2
5	$-\frac{1}{5}$
-4	$\frac{1}{4}$
$\frac{1}{3}$	-3
$-\frac{3}{4}$	$\frac{4}{3}$
$\frac{2}{5}$	$-\frac{5}{2}$

Worked Example: Find an equation of the line perpendicular to $y = 4x + 1$ that passes through $(8, -3)$.

$$\text{Perpendicular gradient} = -\frac{1}{4}. \quad -3 = -\frac{1}{4} \times 8 + c \Rightarrow -3 = -2 + c \Rightarrow c = -1$$

$$y = -\frac{1}{4}x - 1 \quad (\text{or } x + 4y = -4)$$

Worked Example: Show that the lines $4y = 3x - 8$ and $3y + 4x = 6$ are perpendicular.

$$4y = 3x - 8 \Rightarrow y = \frac{3}{4}x - 2, \text{ gradient } \frac{3}{4}. \quad 3y + 4x = 6 \Rightarrow y = -\frac{4}{3}x + 2, \text{ gradient } -\frac{4}{3}.$$

$$\frac{3}{4} \times \left(-\frac{4}{3}\right) = -1, \text{ so the lines are } \mathbf{perpendicular}.$$

Perpendicular Bisectors

The perpendicular bisector of AB goes through the **midpoint** of AB at **right angles** to AB . You need both facts: the midpoint gives a point on the line, the negative reciprocal gives its gradient.

Worked Example: Find an equation of the perpendicular bisector of $A(-1, 2)$ and $B(5, 6)$.

$$\text{Midpoint} = (2, 4). \quad \text{Gradient } AB = \frac{6 - 2}{5 - (-1)} = \frac{2}{3}, \text{ so the perpendicular gradient is } -\frac{3}{2}.$$

$$4 = -\frac{3}{2} \times 2 + c \Rightarrow c = 7 \Rightarrow y = -\frac{3}{2}x + 7 \quad (\text{or } 3x + 2y = 14)$$

Using Gradients to Prove Properties of Shapes

Worked Example: $A(0, 3)$, $B(4, 5)$, $C(6, 1)$ and $D(2, -1)$. Prove that $ABCD$ is a rectangle.

$$m_{AB} = \frac{2}{4} = \frac{1}{2}, \quad m_{DC} = \frac{1 - (-1)}{6 - 2} = \frac{1}{2}, \quad m_{BC} = \frac{1 - 5}{6 - 4} = -2, \quad m_{AD} = \frac{-1 - 3}{2 - 0} = -2.$$

Both pairs of opposite sides are parallel, and $\frac{1}{2} \times (-2) = -1$ so neighbouring sides are perpendicular. Therefore $ABCD$ is a **rectangle**.

Exam Tip: Check a perpendicular gradient by multiplying it by the original: the answer must be exactly -1 . Changing only the sign ($4 \rightarrow -4$) or only flipping ($4 \rightarrow \frac{1}{4}$) are the two

classic errors.

Quick Reference Summary

Fact	Formula / rule
Midpoint	$\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$
Gradient	$m = \frac{y_2-y_1}{x_2-x_1}$
Straight line	$y = mx + c$, y -intercept $(0, c)$
Crossing the x -axis	Put $y = 0$
Parallel lines	Same gradient
In context	Gradient = rate of change; intercept = starting value
Perpendicular lines Higher	$m_1 \times m_2 = -1$